RNS INSTITUTE OF TECHNOLOGY, BENGALURU

Department of Electronics and Communication Engineering

Privacy-Preserving Through-Wall Human Activity Recognition Using WiFi-CSI

Master Technical Guide & Comprehensive Theoretical Reference for Examination Panel

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Project Team: Md Mehar Adil D H (1RN23EC094), Mohith Gowda M (1RN23EC095), Nihar Gowda (1RN23EC106), Sujal L G (1RN23EC169)
Project Guide: Dr. Leena Chandrashekar, Assoc. Prof. | **Project Reviewer:** Dr. Srikanth G N, Assoc. Prof.
Hardware Setup: 2× ESP32-S3 (Dual-core Xtensa LX7 @ 240 MHz, 2.4 GHz HT40, 100 Hz ESP-NOW)

1. Executive Summary & Physical Foundations

This project presents an end-to-end, low-cost through-wall Human Activity Recognition (HAR) and emergency fall detection system using Channel State Information (CSI) extracted from standard Wi-Fi packets transmitted between two ESP32-S3 microcontrollers. Unlike optical cameras that compromise user privacy or wearable devices that suffer from non-compliance, Wi-Fi sensing operates contactlessly, penetrates structural drywall and partitions, and captures zero visual data.

The Multipath Channel Physics: In an indoor environment, transmitted RF waves propagate along multiple paths (direct path, wall reflections, body reflections) before arriving at the receiver antenna. The complex baseband channel response on subcarrier *k* at time *t* is the superposition of *L* paths:

$$H_k(t) = \sum a_i(t) \cdot \exp(-j\,[2\pi\, f_k\, \tau_i(t) + \phi_i(t)]) = H_k^s + H_k^d(t)$$

where *H_k^s* represents the static channel vector (stationary walls, desks, floor) and *H_k^d(t)* represents the dynamic multipath component scattered by the human body. As a person moves, the path length *d_i(t)* changes. At 2.462 GHz (Channel 11), the RF wavelength is $\lambda = c / f \approx$ **12.2 cm**. A body displacement of just 3 cm alters the reflected round-trip path by 6 cm (λ/2), causing a 180° phase flip that turns constructive interference into destructive interference. The measured amplitude fluctuates between |H^s| - |H^d| and |H^s| + |H^d|, generating dynamic Doppler signatures ($f_D = 2v / \lambda$) characteristic of specific human motor actions.

2. ESP32-S3 Hardware & Signal Acquisition Pipeline

The system uses two commodity Espressif ESP32-S3 development boards with external antennas:

Subsystem	Specification	Engineering Rationale
Microcontroller	ESP32-S3-N16R8 (Dual Xtensa LX7 @ 240 MHz)	Low cost, official ESP-IDF CSI driver API support, 8 MB PSRAM.
Transmitter (TX)	6 dBi omnidirectional 2.4 GHz dipole antenna	Uniformly illuminates the indoor sensing area.

Receiver (RX)	8 dBi ALFA directional patch antenna	Concentrates sensitivity toward monitored room, rejecting rear multipath.
Radio Protocol	ESP-NOW Broadcast (Vendor Action Frames)	Connectionless, zero pairing/association overhead, steady 100 packets/s.
PHY Channel	Channel 11 (2.462 GHz primary), HT40 (40 MHz)	HT40 bandwidth doubles subcarriers to 128 HT-LTF for finer multipath.
Packet Rate	100 packets/sec (T = 10 ms period)	Nyquist limit is 50 Hz; captures walking Doppler (~16 Hz) smoothly.

3. Preprocessing & Tensor Standardization Pipeline

Every received packet produces 384 signed 8-bit integers representing 192 complex subcarrier entries. The preprocessing engine transforms raw serial lines into fixed (250 × 90) tensors through six deterministic steps:

Step 1 — Packet Validation: Verify exactly 25 comma-separated CSV fields starting with the CSI_DATA marker and containing 384 finite integers. Corrupted lines are safely discarded.
Step 2 — Amplitude Extraction: Pair integers into in-phase (I) and quadrature (Q) components: **$|H_k| = \sqrt{(I_k^2 + Q_k^2)}$**. Raw phase is discarded because unsynchronized oscillator Carrier Frequency Offset (CFO) and Sampling Frequency Offset (SFO) introduce severe packet-to-packet random slope drift.
Step 3 — Null Subcarrier Elimination: Direct Current (DC) central subcarriers and guard band edge carriers carry zero power and report 0.0. Dropping these prevents artificial edge discontinuities.
Step 4 — Linear Interpolation to Exactly 90 Features: Active subcarriers (~108 to 114) are resampled onto a 90-point uniform grid: x'_j = x_i + (x_{i+1} - x_i) · [(v_j - u_i) / (u_{i+1} - u_i)]. This guarantees fixed dimensions compatible with benchmark standards.
Step 5 — Train-Fitted Normalization: Scale inputs using parameters fitted EXCLUSIVELY on training recordings: x■ = (x - x_min^train) / (x_max^train - x_min^train). Never normalize test data using test set bounds.
Step 6 — Temporal Windowing: Packets are cut into overlapping windows: Length W = 250 packets (2.5 s), Stride S = 50 packets (0.5 s step). Output sample dimension: **[250 × 90]**.

4. Scientific Audit: The Window-Level Data Leakage Discovery

The Leakage Mechanism: In the legacy pipeline, 2,241 windows were generated with stride 50 (80% overlap = 200 packets shared between adjacent windows). The 70/15/15 train/val/test split was performed **at the window level**. Consequently, Window *i* in Train shared 80% identical packets with Window *i+1* in Test! Over 90% of test windows had near-identical sister windows in the training set.

Empirical Proof of Generalization Failure: When evaluating the baseline checkpoint (reporting 97.63% test accuracy on the random split) against 25 new unseen recordings in `live_recordings/`, **accuracy collapsed to 20.55%** (random chance across 5 classes is 20.00%). This proved that prior 98% numbers reflected memorization of room multipath, not generalized activity recognition.

The Leakage-Resistant Recording-Grouped Protocol: We quarantined entire CSV recordings into disjoint Train, Val, and Test subsets BEFORE windowing (strictly 0% window overlap across sets). We enriched the dataset with 25 new recordings, giving **76 total recording sessions** and expanding Falling windows from 43 to 150.

5. Model Architecture Specifications & Mathematical Mechanics

The Input Layout Correction: In PyTorch, `nn.Conv1d` treats dimension 1 as channels. The original repository code reshaped input to [B, 250, 90], treating 250 time packets as unordered channels and convolving along frequency. Permuting to **[B, 90, 250]** treats 90 subcarriers as channels and convolves along time, acting as a learned temporal filter bank. This single physical correction increased test accuracy from 91.10% to 97.63%.

Model Architecture	Params	Latency	Legacy Acc (Mean±SD)	Grouped Acc (Mean±SD)	Key Technical Trait
Baseline MLP	11.58M	1.37 ms	89.91%	46.33%	Fully connected; severe temporal overfitting.
Baseline LeNet	163k	4.00 ms	96.44%	52.41%	2D image CNN; isotropic kernels.
Baseline BiLSTM	80k	1.56 ms	72.11%	41.93%	Pure recurrent; high-frequency noise.
Baseline CNN_GRU	153k	2.76 ms	97.63%	45.28%	1D CNN + unidirectional GRU.
Baseline CNN_BiLSTM	319k	2.39 ms	94.36 ± 2.16%	49.39 ± 3.89%	Original reference; 128 hidden BiLSTM.
Proposed MS_CNN_BiGRU	188k	12.7 ms	97.09 ± 0.95%	50.02 ± 4.63%	Kernels 3,5,7 + SE Attention + BiGRU.
Proposed SE_TCN (Champ)	110k	2.75 ms	97.15 ± 0.61%	58.57 ± 2.64%	Dilated causal convolutions; +9.18% over bas

6. Loss Formulations, Imbalance Ablations & Optimization

Cross-Entropy vs. Focal Loss: Standard Cross-Entropy loss is: L_CE = -log(p_y). When classes are imbalanced (Falling is ~5% of data), the majority classes dominate the loss gradient, causing the model to achieve 58% overall accuracy while detecting only 10.5% of real falls. **Focal Loss** adds a dynamic focusing parameter γ=2.0: **L_Focal = -α_t (1 - p_t)^γ log(p_t)**. When a fall sample has low predicted probability (p_t << 1), the modulating factor (1 - p_t)^2 approaches 1.0, preserving full gradient pressure. Our ablations proved that **Focal Loss tripled Fall Recall from 10.5% to 31.6%**.

The Adam Optimizer Mechanics: Adam maintains running first moments (momentum, β1=0.9) and second moments (uncentered variance, β2=0.999). After bias correction, parameter update is: θ_{t+1} = θ_t - [α / (√(v■_t) + ε)] · m■_t. This assigns adaptive step sizes to each parameter, stabilizing training across convolutional and recurrent layers.

7. Scientific Comparison with Strohmayer & Kampel (ICVS 2023)

Julian Strohmayer and Martin Kampel (ICVS 2023) demonstrated through-wall activity recognition using 2 ESP32 boards transmitting 100 Hz ESP-NOW packets through **4 structural brick walls across an 18-meter distance**, achieving **88.16% ± 4.6%** accuracy on 3 coarse classes (Empty, Walking, Walk+Wave) using an **EfficientNetV2-S** model (20M parameters).

Why higher accuracy on our dataset does NOT prove our model is superior:

1. Our setup uses a single visual partition board at 3–4 meters, not 4 solid brick walls at 18 meters.
2. Our dataset evaluates a harder 5-class problem (including Sitting vs. Standing and Falling).
3. Our genuine contribution over Strohmayer is **extreme edge efficiency**: our SE_TCN model uses only **110,165 parameters** (over **180× smaller** than EfficientNetV2-S), making it viable for direct microcontroller deployment.

8. Master Technical Glossary: Definitions & Worked Examples

Every single technical and theoretical term used in the presentation, repository, and paper, explained with simple real-world examples:

OFDM (Orthogonal Frequency Division Multiplexing)

Meaning: Splits a wide radio channel into many narrow, orthogonal subcarriers sent in parallel.

Real-World Example: *Like a wide highway divided into 128 narrow lanes where cars do not interfere.*

Subcarrier

Meaning: One individual frequency lane within the OFDM channel (spaced 312.5 kHz apart).

Real-World Example: *Subcarrier 0 is at 2.452 GHz, subcarrier 1 at 2.4523125 GHz.*

CSI (Channel State Information)

Meaning: A complex matrix reporting how the wireless channel attenuated and delayed each subcarrier.

Real-World Example: *One packet reports 192 complex numbers: $H_k = I_k + jQ_k$.*

In-Phase (I) & Quadrature (Q)

Meaning: The real (cosine-aligned) and imaginary (sine-aligned) components of the downconverted RF baseband signal.

Real-World Example: *If I = 3 and Q = 4, the complex coefficient is 3 + j4.*

CSI Amplitude

Meaning: The absolute signal strength on subcarrier k: $|H_k| = \sqrt{(I^2 + Q^2)}$.

Real-World Example: *For I=3, Q=4: Amplitude = $\sqrt{(9 + 16)} = 5.0$.*

CSI Phase

Meaning: The angle of the signal vector: $\phi = \text{atan2}(Q, I)$.

Real-World Example: *atan2(4, 3) ≈ 53.1°. Discarded because oscillator drift adds random offsets.*

Multipath Fading

Meaning: Signal copies arriving over different paths with different delays, adding constructively or destructively.

Real-World Example: *Walking through the room causes peaks and nulls as waves reinforce or cancel.*

RF Wavelength (λ)

Meaning: Physical distance spanned by one full radio wave cycle: $\lambda = c / f$.

Real-World Example: *At 2.462 GHz: $\lambda = (3\times10^8) / (2.462\times10^9) \approx 12.2$ cm.*

RSSI

Meaning: Received Signal Strength Indicator: a single scalar measuring total received power.

Real-World Example: *-55 dBm. Averages all subcarriers together, losing spatial multipath details.*

ESP-NOW

Meaning: Espressif's connectionless Wi-Fi MAC action frame protocol.

Real-World Example: *TX broadcasts packets directly to RX every 10 ms without router pairing.*

HT40

Meaning: High Throughput 40 MHz channel bandwidth (IEEE 802.11n).

Real-World Example: *Doubles bandwidth from 20 MHz, giving 128 HT-LTF subcarriers for finer resolution.*

Null Subcarrier

Meaning: Guard band edge carriers and DC central carrier carrying zero energy.

Real-World Example: *Reported amplitudes [0, 5, 6, 0, 7] -> zeros dropped to keep active [5, 6, 7].*

Resampling (Linear Interpolation)

Meaning: Mapping variable active subcarriers onto a fixed 90-point grid by joining dots with straight lines.

Real-World Example: *Maps ~108 active carriers to exactly 90 features so tensor size is constant.*

Min-Max Normalization

Meaning: Rescaling values into [0, 1] using training set bounds: (x - min) / (max - min).

Real-World Example: *Range [1.0, 25.88]: value 13.44 becomes (13.44 - 1)/(24.88) = 0.50.*

Temporal Window

Meaning: A fixed block of consecutive packets treated as one model input sample.

Real-World Example: *250 consecutive packets ≈ 2.5 seconds of motion at 100 packets/s.*

Stride & Overlap

Meaning: Stride is how far the window advances; overlap is packets shared with adjacent windows.

Real-World Example: *Window=250, Stride=50 -> Overlap = 250 - 50 = 200 packets (80% shared).*

Data Leakage

Meaning: Information from the test set contaminating training, creating artificially inflated accuracy.

Real-World Example: *Window 0-249 in train and window 50-299 in test share 200 identical packets.*

Conv1D

Meaning: Slides a small filter along a 1D sequence computing a learned weighted sum.

Real-World Example: *Filter [1, 0, -1] on [2, 4, 6] produces 1·2 + 0·4 + (-1)·6 = -4.*

Receptive Field

Meaning: How many original input time steps contribute to a single deep activation.

Real-World Example: *After Conv1 (k=5) + Conv2 (k=3) + Pool (2), each activation sees ~8 packets (~80 ms).*

BiLSTM

Meaning: Two LSTMs: one reads time forward (0->125), one backward (125->0).

Real-World Example: *Forward summarizes motion ending; backward summarizes from rest back to start.*

BiGRU

Meaning: Bidirectional Gated Recurrent Unit: uses 2 gates (reset/update) instead of 3 gates.

Real-World Example: *Saves 33% recurrent parameters over LSTM with equal memory retention.*

Dilated Convolution

Meaning: Convolution where filter taps have gaps (dilation factor d).

Real-World Example: *Dilation d=2 on filter [a, b, c] samples steps t-2, t, t+2, expanding field exponentially.*

SE Attention

Meaning: Squeeze-and-Excitation: learns to dynamically scale informative channels.

Real-World Example: *Assigns weights in [0, 1] to boost high-SNR subcarriers and suppress fading nulls.*

Dropout

Meaning: Randomly zeros a fraction p of activations during training (disabled at test time).

Real-World Example: *p=0.15: 15% of features set to zero, preventing neuron co-adaptation.*

Logits

Meaning: Raw unconstrained real-valued scores output by the final linear layer.

Real-World Example: *Logits [-1.2, 0.5, 3.4] are converted by softmax into probabilities.*

Softmax

Meaning: Converts logits z into posterior probabilities that sum to 1.0: exp(z_c) / ∑exp(z).

Real-World Example: *Logits [0, 2] -> exp(0)=1, exp(2)=7.39 -> Probabilities = [0.12, 0.88].*

Cross-Entropy Loss

Meaning: Negative log-likelihood of the true class: -log(p_y).

Real-World Example: *If true class prob = 0.90, loss = -ln(0.90) = 0.105; if prob = 0.10, loss = 2.30.*

Focal Loss

Meaning: Downweights easy examples with factor (1 - p_t)^γ to focus on hard minority classes.

Real-World Example: *Tripled Fall Recall from 10.5% to 31.6% by forcing network to learn rare falls.*

Adam Optimizer

Meaning: Maintains moving averages of gradient (momentum) and squared gradient (variance).

Real-World Example: *Gives each parameter an adaptive learning rate; ideal for hybrid CNN-RNN models.*

Early Stopping

Meaning: Halting training when validation metric stops improving for patience epochs.

Real-World Example: *Patience 6: stops if validation Macro-F1 does not improve for 6 epochs, saving best model.*

Balanced Accuracy

Meaning: Unweighted average of recall across all classes: (1/C) ∑ Recall_c.

Real-World Example: *Unlike raw accuracy, balanced accuracy is unaffected by majority Empty samples.*

Macro F1-Score

Meaning: Harmonic mean of precision and recall averaged equally across all 5 classes.

Real-World Example: *A model with 99% accuracy on Empty but 0% on Falling will have poor Macro-F1 (~0.70).*

Confusion Matrix

Meaning: Table where rows are true classes and columns are predicted classes.

Real-World Example: *Diagonal entries are correct classifications; off-diagonals are mistakes.*